Linear Algebra

Author: Isaac FEI

Preface

Linear algebra is a fascinating field of mathematics that offers a powerful set of tools for modeling and analyzing real-world problems. From designing computer graphics to solving systems of equations, linear algebra has numerous applications in science, engineering, and technology. With its combination of abstract theory and practical applications, linear algebra offers a unique opportunity to explore the interplay between mathematics and the real world.

The main structure of this book follows Sheldon Axler's *Linear Algebra Done Right* [1], which emphasizes understanding vector spaces and linear transformations before introducing matrices and determinants. However, I believe that more examples and applications related to matrices can help students better understand the subject, and therefore I aim to supplement Axler's approach with additional content in this area.

This book covers a wide range of topics, starting from basic vector space theory and linear systems to more advanced topics such as eigenvalues, singular values, and applications in data science and machine learning. By providing both theoretical and practical insights, I hope to offer readers a comprehensive and integrated understanding of the subject.

Contents

Chapter 1 Finite-Dimensional Vector Spaces **1**

1.1 Bases 1

Chapter 2 Linear Maps **2**

2.1 Null Spaces and ranges 2

2.2 Projections 2

Chapter 3 Polynomials **4**

3.1 Polynomial with Real Coefficients 4

Chapter 4 Matrix Algebra **5**

4.1 LU Decomposition 5

4.2 LU Decomposition with Partial Pivoting 5

Chapter 5 Eigenvalues and Eigenvectors **6**

5.1 Invariant Subspaces 6

5.2 Upper Triangular Matrices 6

5.3 Diagonal Matrices 10

Chapter 6 Inner Product Spaces **14**

6.1 Inner Products 14

6.2 Norms 14

6.3 Orthonormal Bases 18

6.4 Gram-Schmidt Process 20

6.5 QR Decomposition 22

6.6 Upper Triangular Matrices With Respect to Orthonormal Bases 25

6.7 Orthogonal Projections 26

6.8 Linear Functionals and Adjoints 30

References **33**

Index **34**

Chapter 1 Finite-Dimensional Vector Spaces

1.1 Bases

Theorem 1.1.1

Every linearly independent list of vectors in a finite-dimensional vector space can be extended to a basis. ♡

Proof ■

Chapter 2 Linear Maps

2.1 Null Spaces and ranges

Theorem 2.1.1

Let $T : V \to W$ be a linear map. Then T is injective if and only if $\ker T = \{\mathbf{0}\}$. ♡

Proof ■

Corollary 2.1.1

Let $T : V \to W$ be a linear map. Then T is injective if and only if $(T\mathbf{v}_1, \ldots, T\mathbf{v}_n)$ is linearly independent in W as long as $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is linearly independent in V. Mathematically,

$$(\mathbf{v}_1, \ldots, \mathbf{v}_n) \text{ is linearly independent in } V \implies (T\mathbf{v}_1, \ldots, T\mathbf{v}_n) \text{ is linearly independent in } W$$ ♡

Proof ■

Corollary 2.1.2

Let $T : V \to W$ be a linear map where V is finite-dimensional. Suppose $B = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis for V. Then T is injective if and only if $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is linearly independent. ♡

Proof ■

2.2 Projections

Definition 2.2.1

*Let V be a vector space. We say a linear operator $P \in \mathcal{L}(V)$ is a **projection** if it satisfies*

$$P^2 = P$$

♣

Proposition 2.2.1

Let V be a finite-dimensional vector space, and $P \in \mathcal{L}(V)$ a projection. Suppose $U = \operatorname{im} P$ and $W = \ker P$. Then we have

1. *$V = U \oplus W$, and*
2. *$P\mathbf{v} = \mathbf{u}\ \forall \mathbf{v} \in V$ provided that $\mathbf{v} = \mathbf{u} + \mathbf{w}$ where $\mathbf{u} \in U$ and $\mathbf{w} \in W$.*

In particular, $P\mathbf{u} = \mathbf{u}\ \forall \mathbf{u} \in U$. ♠

Proof ■

Proposition 2.2.2

Let V be a finite-dimensional vector space. Suppose that $V = U \oplus W$. Let a linear operator

$P \in \mathcal{L}(V)$ be defined by

$$P\mathbf{v} = \mathbf{u} \quad \forall \mathbf{v} \in V$$

where $\mathbf{v} = \mathbf{u} + \mathbf{w}$, $\mathbf{u} \in U$ and $\mathbf{w} \in W$. (Notice that P is well-defined since each $\mathbf{v} \in V$ can be uniquely written as $\mathbf{v} = \mathbf{u} + \mathbf{w}$ where $\mathbf{u} \in U$ and $\mathbf{w} \in W$.) Then P is in fact a projection. Moreover,

$$U = \operatorname{im} P \qquad\qquad W = \ker P$$

Proof ■

Proposition 2.2.3

Suppose V is a finite-dimensional vector space and $V = U \oplus W$. For each $\mathbf{v} \in V$, write $\mathbf{v} = \mathbf{u} + \mathbf{w}$ where $\mathbf{u} \in U$ and $\mathbf{w} \in W$. Let $P_U, P_W \in \mathcal{L}(V)$ be defined by

$$P_U\mathbf{v} = \mathbf{u} \qquad\qquad P_W\mathbf{v} = \mathbf{w}$$

(We know from Proposition 2.2.2 that P_U and P_W are projections.) We have

$$P_U + P_W = I$$

Proof The conclusion of this proposition is evident since it is clear from the definition that

$$\mathbf{v} = \mathbf{u} + \mathbf{w} = P_U(\mathbf{v}) + P_W(\mathbf{v})$$

for each $\mathbf{v} \in V$ with $\mathbf{v} = \mathbf{u} + \mathbf{w}$ where $\mathbf{u} \in U$ and $\mathbf{w} \in W$. ■

Chapter 3 Polynomials

3.1 Polynomial with Real Coefficients

Theorem 3.1.1

Let $p(x)$ be a nonconstant polynomial with real coefficients. Then it has a unique factorization (except for the order of factors) of the form

$$p(x) = c(x - \lambda_1) \cdots (x - \lambda_k)(x^2 + \alpha_1 x + \beta_1) \cdots (x^2 + \alpha_l x + \beta_l)$$

where $k + m \geq 1$ and $\lambda_j, \alpha_j, \beta_j \in \mathbb{R}$ with $\alpha_j^2 < 4\beta_j$. ♡

Proof ■

Chapter 4 Matrix Algebra

4.1 LU Decomposition

4.2 LU Decomposition with Partial Pivoting

Sometimes, even when A is a nonsingular matrix, it may not admit an LU decomposition. To fix this, we need to first rearrange the rows of A in some way, and then it is guaranteed to have an LU decomposition if it is nonsingular.

The rearrangement of rows of A can be done by multiplying a permutation matrix P on the left of A. In this case, the decomposition can be written as

$$PA = LU$$

Algorithm 1: LU Decomposition with Partial Pivoting

Input: $A \in \mathbb{F}^{n \times n}$
Output: L, U, P

```
U ← A;
L ← I_n;
P ← I_n;
for i = 1, ..., n do
    k ← arg max_k |U_{k,i}| s.t. k ≥ i;
    // U_{k,i} is a pivot entry
    if U_{k,i} = 0 then
        throw error Singular matrix does not admit an LU decomposition.
    end
    if i ≠ k then
        swap row_i U and row_k U;
        swap row_i P and row_k P;
        swap row_i L and row_k L;
        swap col_i L and col_k L;
    end
    for j = i + 1, ..., n do
        L_{j,i} ← U_{j,i} / U_{i,i};
        row_j U ← row_j U − L_{j,i} × row_i U;
    end
end
```

Chapter 5 Eigenvalues and Eigenvectors

5.1 Invariant Subspaces

Example 5.1 Consider the linear operator on $\mathbb{R}^2$ whose matrix representation with respect to the standard basis is given by

$$[T] = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \qquad \sin\theta \neq 0$$

Geometrically, T rotates any point (x, y) on the 2-dimensional plane counterclockwise through an angle θ about the origin. But since we additionally require that $\sin\theta \neq 0$, the point cannot land at (x, y) or $(-x, -y)$ after rotation. That is, it cannot land on the *line* passing through the origin and (x, y), which is precisely $\operatorname{span}\{(x, y)^\top\}$. Therefore, we see that T has no eigenvalues.

✍ **Exercise 5.1** Show by computation that the linear operator T in Example 5.1 indeed does not have any eigenvalues.

5.2 Upper Triangular Matrices

Theorem 5.2.1

Any operator on a finite-dimensional nonzero complex vector space has an eigenvalue. ♡

Proof ■

Proposition 5.2.1

Let V be a vector space and $T \in \mathcal{L}(V)$. Suppose $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis for V, then the following are equivalent:

1. *$[T]$ is an upper triangular matrix with respect to the basis $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$.*
2. *$T\mathbf{v}_k \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ for all $k = 1, \ldots, n$.*
3. *$\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ is invariant under T for all $k = 1, \ldots, n$.* ♠

Proof ■

Theorem 5.2.2

Let V be a complex vector space, and $T \in \mathcal{L}(V)$. Then there exists some basis B for V such that $[T]_B$ is upper triangular. ♡

Remark In fact, as we shall see in Section 6.6, we can find an *orthonormal* basis B such that $[T]_B$ is upper triangular, which is a stronger version of this theorem.

To prove this theorem, we first use Theorem 5.2.1 to find an eigenvalue λ for T. After that, we construct a T-invariant subspace $\operatorname{im}(T - \lambda I)$, which has smaller dimension of V. Then we may prove this by mathematical induction.

Proof We shall prove by mathematical induction. The hypothesis $P(n)$ is stated as

$$P(n) : \text{This theorem holds for } \dim V \leq n.$$

(Base Case: $\dim V = 1$) If V has dimension 1, then this theorem holds trivially since $[T]$ consists of one entry, which clearly upper triangular.

(Inductive Step) Assume $P(k)$ ($k \geq 1$) holds, we need to show that $P(k+1)$ also holds. Note that it suffices to prove this theorem holds for $\dim V = k+1$. Firstly, by Theorem 5.2.1, T has an eigenvalue λ. Consider the subspace

$$U = \operatorname{im}(T - \lambda I)$$

We claim that U is invariant under T. Pick any $\mathbf{u} \in U$, we need to show $T\mathbf{u} \in U$. Since $\mathbf{u} \in U$, there exists some $\mathbf{v} \in V$ such that $\mathbf{u} = (T - \lambda I)\mathbf{v}$. It then follows that

$$T\mathbf{u} = T(T - \lambda I)\mathbf{v} = (T - \lambda I)T\mathbf{v} = (T - \lambda I)(T\mathbf{v}) \in U$$

This shows U is indeed invariant under T, which implies that the restricted map $T|_U$ is an operator on U.

Note that $\dim U < \dim V = k+1$, which implies $\dim U \leq k$. Now, we may apply the induction hypothesis $P(k)$ on U. It follows that the operator $[T|_U]_{B'}$ is upper triangular for some basis

$$B' = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$$

of subspace U.

We then extend B' to a basis B for V by adding some vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{k-m+1}$. In this case,

$$B = (\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{v}_1, \ldots, \mathbf{v}_{k-m+1})$$

One may conjecture that $[T]_B$ is an upper triangular matrix, which is indeed true, and we shall now prove this by verifying statement 2 in Proposition 5.2.1. Since U is invariant under T, it is clear

$$T\mathbf{u}_j \in \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_j) \quad \forall j = 1, \ldots, m$$

As for $\mathbf{v}_j$, we have

$$T\mathbf{v}_j = (T - \lambda I)\mathbf{v}_j + \lambda \mathbf{v}_j$$

Note that $(T - \lambda I)\mathbf{v}_j \in U$. Thus,

$$(T - \lambda I)\mathbf{v}_j \in \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_j, \mathbf{v}_j) \subset \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_j, \mathbf{v}_1, \ldots, \mathbf{v}_j) \quad \forall j = 1, \ldots, k-m+1$$

Therefore, $[T]_B$ is indeed upper triangular. This completes the proof. ■

If somehow we know that the matrix representation of a linear operator T is upper triangular for some basis B. Then it becomes extremely easy to check whether it is invertible. In a word, T is invertible if and only if all diagonal entries of matrix $[T]_B$ are nonzero as stated in Proposition 5.2.2.

To make the proof of Proposition 5.2.2 cleaner, we fist prove the following lemma.

Lemma 5.2.1

Let V be a finite-dimensional vector space, and $T \in \mathcal{L}(V)$. Suppose $[T]_B$ is an upper triangular matrix for some basis $B = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ of V. Then $(T\mathbf{v}_1, \ldots, T\mathbf{v}_n)$ is linearly independent if all diagonal entries of $[T]_B$ are nonzero. ♡

Proof We shall prove by induction on $\dim V = n$.

(Base Case: $n = 1$) In this case, suppose $B = (\mathbf{v})$, and then we have $[T]_B = [\lambda]$ for some nonzero λ. It follows that $T\mathbf{v} = \lambda\mathbf{v} \neq 0$. The list $(T\mathbf{v})$ is clearly linearly independent since it consists only one nonzero vector.

(Inductive Step) Assume this lemma holds for $\dim V = k$ ($k \geq 1$). Now, suppose $\dim V = k+1$ and the basis $B = (\mathbf{v}_1, \ldots, \mathbf{v}_{k+1})$. Write

$$[T]_B = \begin{bmatrix} \lambda_1 & a_{1,2} & a_{1,3} & \cdots & a_{1,k+1} \\ & \lambda_2 & a_{2,3} & \cdots & a_{2,k+1} \\ & & \ddots & \ddots & \vdots \\ & & & \lambda_k & a_{k,k+1} \\ 0 & & & & \lambda_{k+1} \end{bmatrix} \tag{5.1}$$

The given condition implies that $\lambda_j \neq 0 \ \forall j = 1, \ldots, k+1$.

Let

$$U = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$$

We know from statement 3 in Proposition 5.2.1 that U is invariant under T. Hence, $T|_U$ is an operator on U. Let $B' = (\mathbf{v}_1, \ldots, \mathbf{v}_k)$. It is clear that

$$[T|_U]_{B'} = \begin{bmatrix} \lambda_1 & a_{1,2} & \cdots & a_{1,k} \\ & \lambda_2 & \cdots & a_{2,k} \\ & & \ddots & \vdots \\ 0 & & & \lambda_k \end{bmatrix}$$

It then follows from the induction hypothesis that $(T\mathbf{v}_1, \ldots, T\mathbf{v}_k)$ is linearly independent.

The last column of (5.1) gives

$$T\mathbf{v}_{k+1} = a_{1,k+1}\mathbf{v}_1 + \cdots + a_{k,k+1}\mathbf{v}_k + \lambda_{k+1}\mathbf{v}_{k+1} \tag{5.2}$$

Set

$$c_1 T\mathbf{v}_1 + \cdots + c_k T\mathbf{v}_k + c_{k+1} T\mathbf{v}_{k+1} = \mathbf{0} \tag{5.3}$$

Plugin (5.2),

$$c_1 T\mathbf{v}_1 + \cdots + c_k T\mathbf{v}_k + c_{k+1}(a_{1,k+1}\mathbf{v}_1 + \cdots + a_{k,k+1}\mathbf{v}_k) + c_{k+1}\lambda_{k+1}\mathbf{v}_{k+1} = \mathbf{0} \tag{5.4}$$

Note that all terms expect for the last one on the left-hand side of (5.4) is in $\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$. Hence, (5.4) yields $c_{k+1} = 0$. Then (5.4) itself reduces to

$$c_1 T\mathbf{v}_1 + \cdots + c_k T\mathbf{v}_k = \mathbf{0}$$

This implies $c_1 = \cdots = c_k = 0$ since we have already shown that $(T\mathbf{v}_1, \ldots, T\mathbf{v}_k)$ is linearly independent. In summary, (5.3) finally yields that $c_1 = \cdots = c_{k+1} = 0$, which means $(T\mathbf{v}_1, \ldots, T\mathbf{v}_{k+1})$ is linearly independent. This completes the proof. ■

Proposition 5.2.2

Let T be a linear operator on V. Suppose $[T]_B$ is an upper triangular matrix for some basis B of V. Then T is invertible if and only if all diagonal entries of $[T]_B$ are nonzero. ♠

Proof Suppose $\dim V = n$ and $B = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$. Write

$$[T]_B = \begin{bmatrix} \lambda_1 & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ & \lambda_2 & a_{2,3} & \cdots & a_{2,n} \\ & & \lambda_3 & \cdots & a_{3,n} \\ & & & \ddots & \vdots \\ 0 & & & & \lambda_n \end{bmatrix}$$

We have

$$T\mathbf{v}_j = a_{1,j}\mathbf{v}_1 + \cdots + a_{j-1,j}\mathbf{v}_{j-1} + \lambda_j \mathbf{v_j} \tag{5.5}$$

Recall Corollary 6.4.1, and what we need to prove will become that $(T\mathbf{v}_1, \ldots, T\mathbf{v}_n)$ is linearly independent if and only if $\lambda_j \neq 0 \; \forall j = 1, \ldots, n$.

(Sufficiency) This is proved by Lemma 5.2.1.

(Necessity) For the necessity, we shall prove by *contrapositive*. Assume $\lambda_j = 0$ for some j. If $j = 1$, then (5.5) yields $T\mathbf{v}_1 = \mathbf{0}$, which implies $(T\mathbf{v}_1, \ldots, T\mathbf{v}_n)$ is linearly dependent. If $j \geq 2$, then it follows from (5.5) that

$$T\mathbf{v}_j \in \operatorname{span}(T\mathbf{v}_1, \ldots, T\mathbf{v}_{j-1})$$

This again implies $(T\mathbf{v}_1, \ldots, T\mathbf{v}_n)$ is linearly dependent. ■

Another application of Proposition 5.2.2 is to prove the following theorem, which states that the eigenvalues of T consists precisely of the diagonal entries of $[T]_B$ provided that it is an upper triangular matrix.

Theorem 5.2.3

Let V be a vector space and $T \in \mathcal{L}(V)$. If $[T]_B$ is upper triangular for some basis B of V, then the eigenvalues of T consists precisely of diagonal entries of $[T]_B$. ♡

Remark This theorem in fact says two things. One is that all diagonal entries of $[T]_B$ must be eigenvalues. And the other one is that if λ is an eigenvalue of T, then it must appear at the diagonal of $[T]_B$.

Proof Write

$$[T]_B = \begin{bmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

We first show that each λ_j is an eigenvalue of T. It is clear that

$$[T - \lambda_j I]_B = \begin{bmatrix} \lambda_1 - \lambda_j & & & & * \\ & \ddots & & & \\ & & \lambda_j - \lambda_j = 0 & & \\ & & & \ddots & \\ 0 & & & & \lambda_n - \lambda_j \end{bmatrix}$$

Since $[T - \lambda_j I]_B$ is upper triangular and there exists a zero entry on its diagonal, it then follows from Proposition 5.2.2 that $T - \lambda_j I$ is not invertible, which implies λ_j is an eigenvalue of T.

Let λ be an eigenvalue of T, we also need to show that λ appears on the diagonal of $[T]_B$. We have

$$[T-\lambda I]_B = \begin{bmatrix} \lambda_1 - \lambda & & * \\ & \ddots & \\ 0 & & \lambda_n - \lambda \end{bmatrix}$$

Note that $[T-\lambda I]_B$ is upper triangular and $T-\lambda I$ is not invertible since λ is an eigenvalue of T. Thus, again by Proposition 5.2.2, we conclude that there is some zero entry on the diagonal of $[T-\lambda I]_B$. That is, $\lambda_j = \lambda$ for some j, which implies λ indeed appears on the diagonal of $[T]_B$. ■

Let B_1 and B_2 be two bases of V, and T a linear operator. Suppose $[T]_{B_1}$ and $[T]_{B_2}$ are both upper triangular matrices. Now, we know that the **sets** of diagonal entries of $[T]_{B_1}$ and $[T]_{B_2}$ must the same, which both consist of **distinct** eigenvalues of T.

But is it possible that the occurrence of the same eigenvalue λ on the diagonal differ from each other? For example, if λ appears k times on the diagonal of $[T]_{B_1}$, then must λ appear also k times on the diagonal of $[T]_{B_2}$?

The answer is yes. But to prove this we need to introduce the concept of **characteristic polynomials**, which will be covered in a later chapter.

5.3 Diagonal Matrices

For an operator T on a real vector space, it may not have any eigenvalues, as we have seen in Example 5.1. Hence, it is possible that no 1-dimensional subspace exists that is invariant under T.

However, the following theorem states that a T-invariant subspace with dimension 1 or 2 is always possible to find.

Theorem 5.3.1

Let V be a finite-dimensional nonzero vector space over $\mathbb{R}$, and $T \in \mathcal{L}(V)$. Then there exists a subspace $U \leq V$ such that U is invariant under T and $\dim U = 1$ or $\dim U = 2$. ♡

Proof Suppose $\dim V = n$. Pick a nonzero vector $\mathbf{v} \neq \mathbf{0}$ in V. Consider the following list of vectors

$$(\mathbf{v}, T\mathbf{v}, \ldots, T^n\mathbf{v})$$

Notice that it is clearly linearly dependent since it consists of $n+1$ vectors. Then there exist some real numbers a_j's that are not all zeros such that

$$a_0\mathbf{v} + a_1T\mathbf{v} + \cdots a_nT^n\mathbf{v} = \mathbf{0}$$

Let a_m $(m \leq n)$ be the last one among all a_j's that is nonzero. Then the equation above reduces to

$$a_0\mathbf{v} + a_1T\mathbf{v} + \cdots a_mT^m\mathbf{v} = \mathbf{0} \quad a_m \neq 0 \tag{5.6}$$

Define a polynomial

$$p(x) = a_0x + a_1x + \cdots a_mx^m$$

Recall Theorem 3.1.1, which says that we can factorization $p(x)$ into

$$p(x) = c(x-\lambda_1)\cdots(x-\lambda_k)(x^2+\alpha_1x+\beta_1)\cdots(x^2+\alpha_lx+\beta_l)$$

where $c \neq 0$ is some constant and α_j, β_j satisfy $\alpha_j^2 < 4\beta_j$. Apply p to T, and it then follows from (5.6) that

$$(T - \lambda_1 I) \cdots (T - \lambda_k I)(T^2 + \alpha_1 T + \beta_1 I) \cdots (T^2 + \alpha_l T + \beta_l I)\mathbf{v} = \mathbf{0} \tag{5.7}$$

Recall $\mathbf{v} \neq 0$. Then (5.7) implies that the linear map composition on the left-hand side is not injective, which further implies there exists at least one linear map in this composition that is not injective. This leads to two cases.

(Case 1) If $T - \lambda_j I$ is not injective for some j, then λ_j is clearly an eigenvalue of T. Let $\mathbf{u}$ be its corresponding eigenvector, and let $U = \operatorname{span}(\mathbf{u})$. It is clear that U has dimension 1 and is invariant under T.

(Case 2) Suppose $T^2 + \alpha_j T + \beta_j I$ is not injective for some j. There exists $\mathbf{u} \neq \mathbf{0}$ such that

$$(T^2 + \alpha_j T + \beta_j I)\mathbf{u} = \mathbf{0} \tag{5.8}$$

Notice that $\operatorname{span}(\mathbf{u}, T\mathbf{u})$ clearly has dimension 1 or 2. (In fact, it has dimension 2. But we do not need to know this for sure in the proof of this theorem. See the note after the proof.) Now, we show that $\operatorname{span}(\mathbf{u}, T\mathbf{u})$ is invariant under T. Any vector in $\operatorname{span}(\mathbf{u}, T\mathbf{u})$ can be written as

$$c_1\mathbf{u} + c_2 T\mathbf{u}$$

for some real numbers c_1 and c_2. We have

$$T(c_1\mathbf{u} + c_2 T\mathbf{u}) = c_1 T\mathbf{u} + c_2 T^2\mathbf{u} = -c_2\beta_j\mathbf{u} + (c_1 - c_2\alpha_j)T\mathbf{u} \in \operatorname{span}(\mathbf{u}, T\mathbf{u})$$

where the second last equality follows form (5.8). This implies $\operatorname{span}(\mathbf{u}, T\mathbf{u})$ is indeed invariant under T. ■

Note *As we have commented in the proof,* $\operatorname{span}(\mathbf{u}, T\mathbf{u})$ *actually has dimension* 2*, which is equivalent to say* $\mathbf{u}$ *and* $T\mathbf{u}$ *are linearly independent. To see this, we assume they are linearly dependent and hence, there exists some real number* a *such that*

$$T\mathbf{u} = a\mathbf{u}$$

Substitute $a\mathbf{u}$ *for* $T\mathbf{u}$ *in the equation* (5.8) *for several times, and we will obtain*

$$(a^2 + a\alpha_j + \beta_j)\mathbf{u} = \mathbf{0}$$

Recall $\mathbf{u} \neq \mathbf{0}$*, which implies*

$$a^2 + a\alpha_j + \beta_j = 0$$

But this equation has no real solutions since $\alpha_j^2 < \beta_j$*. Therefore, there does not exist a real* a *such that* $T\mathbf{u} = a\mathbf{u}$*. Hence,* $\dim \operatorname{span}(\mathbf{u}, T\mathbf{u}) = 2$.

Theorem 5.3.2

Let V *be an odd-dimensional real vector space. Then any linear operator* $T \in \mathcal{L}(V)$ *has an eigenvalue.* ♡

Proof Suppose $\dim V = 2n - 1$ ($n \in \mathbb{N}^*$). We shall prove by induction on n. Let the induction hypothesis $P(n)$ be that

$$P(n) : \text{This theorem holds for } \dim V = 2n - 1.$$

(Base Case) Let $n = 1$, then $\dim V = 1$. It is clear that T has an eigenvalue in this case. Hence,

$P(1)$ holds.

(Inductive Step) Assume $P(k)$ holds, we shall prove $P(k+1)$. It follows from Theorem 5.3.1 that there exists a subspace $U \leq V$ such that U is invariant under T and $\dim U = 1$ or $\dim U = 2$. If $\dim U = 1$, then it is clear that T has an eigenvalue whose eigenspace *contains* U.

We now consider the scenario where $\dim U = 2$. First, we notice that there exists a subspace W such that

$$V = U \oplus W$$

Notice that $\dim W = 2k - 1$ since $\dim V = 2k + 1$ and $\dim U = 2$. It is tempting to apply the induction hypothesis to W. But unfortunately, there is no guarantee that W is invariant under T!.

Hence, instead of finding an eigenvalue of T directly, we are going to first composite T with a projection so that W is invariant under the composition map, and then obtain an eigenvalue for that composition map. For each $\mathbf{v} \in V$ with $\mathbf{v} = \mathbf{u} + \mathbf{w}$ ($\mathbf{u} \in U$, $\mathbf{w} \in W$), define linear operators

$$P_U \mathbf{v} = \mathbf{u} \qquad\qquad P_W \mathbf{v} = \mathbf{w}$$

We know from Proposition 2.2.2 that both P_U and P_W are projections. For the time being, we will only make use of P_W. (We will need P_U later.) Notice that W is clearly invariant under the composition $P_W T$. Then applying the induction hypothesis $P(k)$, we may conclude that there exists an eigenvalue λ for $P_W T$. Let $\mathbf{w}$ be one of its corresponding eigenvectors.

We will now prove λ is also an eigenvalue of T by showing that the linear operator $T - \lambda I$ is not injective. To do so, we will show that $T - \lambda I$ maps some subspace into a subspace with smaller dimension. Consider the subspace

$$U \oplus \operatorname{span}(\mathbf{w})$$

(Recall $\mathbf{w}$ is an eigenvector of $P_W T$ corresponding to λ.) The reason why we can write $\oplus$ here is that $\mathbf{w} \in W$ ($\mathbf{w} \neq \mathbf{0}$) and hence $\mathbf{w} \notin U$. We claim that $T - \lambda I$ maps $U \oplus \operatorname{span}(\mathbf{w})$ into U. To see this, we pick an arbitrary vector in $U \oplus \operatorname{span}(\mathbf{w})$, which can be written as $\mathbf{u} + a\mathbf{w}$ ($\mathbf{u} \in U$, $a \in \mathbb{R}$), we have

$$(T - \lambda I)(\mathbf{u} + a\mathbf{w}) = (T - \lambda I)\mathbf{u} + aT\mathbf{w} - a\lambda\mathbf{w} \tag{5.9}$$

Notice that

$$\lambda\mathbf{w} = P_W T\mathbf{w} \tag{5.10}$$

(Recall that we have defined P_U previously, and we will use it here.) And it follows from Proposition 2.2.3 that

$$T\mathbf{w} = (P_U + P_W)T\mathbf{w} \tag{5.11}$$

Plug (5.10) and (5.11) into (5.9), we obtain

$$\begin{aligned}(T - \lambda I)(\mathbf{u} + a\mathbf{w}) &= (T - \lambda I)\mathbf{u} + a(P_U + P_W)T\mathbf{w} - aP_W T\mathbf{w} \\ &= (T - \lambda I)\mathbf{u} + aP_U T\mathbf{w} \end{aligned} \tag{5.12}$$

Recall U is invariant under T. Hence, $(T - \lambda I)\mathbf{u} \in U$. Moreover, since P_U projects every vector into U, it is clear $aP_U T\mathbf{w} \in U$. Therefore, (5.12) implies that

$$(T - \lambda I)(\mathbf{u} + a\mathbf{w}) \in U$$

This shows that $T - \lambda I$ maps $U + \operatorname{span}(\mathbf{w})$ into U. Since $U + \operatorname{span}(\mathbf{w})$ clearly has larger dimension

than that of U (actually, it is 1 dimension larger), $T - \lambda I$ is not injective. Therefore, λ is indeed an eigenvalue of T. This completes the proof. ■

Chapter 6 Inner Product Spaces

6.1 Inner Products

Definition 6.1.1

An inner product $\langle \cdot, \cdot \rangle$ on a vector space V over $\mathbb{F}$ is a mapping

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$$

with the following properties:

1. *(Positive-Definiteness)* $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0 \quad \forall \mathbf{v} \in V$ *and* $\langle \mathbf{v}, \mathbf{v} \rangle = 0 \iff \mathbf{v} = \mathbf{0}$.
2. *(Conjugate Symmetry)* $\langle \mathbf{v}, \mathbf{w} \rangle = \overline{\langle \mathbf{w}, \mathbf{v} \rangle} \quad \forall \mathbf{v}, \mathbf{w} \in V$.
3. *(Linearity in the Second Slot)* $\langle \mathbf{v}, a\mathbf{w} + b\mathbf{u} \rangle = a \langle \mathbf{v}, \mathbf{w} \rangle + b \langle \mathbf{v}, \mathbf{u} \rangle$ *for all* $a, b \in \mathbb{F}$ *and for all* $\mathbf{v}, \mathbf{w}, \mathbf{u} \in V$.

*We say V is an **inner product space** if there is an inner product defined on it.* ♣

Note *In most mathematical contexts, a slightly different definition is adopted where the inner product is assumed linear in the first slot, which in fact follows the original definition of inner products. However, as mentioned before, the reason why we require the inner product to be linear in the second slot is to match the notation of Hermitian transpose $\mathbf{x}^H\mathbf{y}$.*

Remark Note that we do not assume V is finite-dimensional in this definition. But in most cases, we shall confine ourselves only to finite-dimensional inner product spaces.

6.2 Norms

Definition 6.2.1

Let V be an inner product space. The norm of a vector $\mathbf{v} \in V$ is defined as

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$$

♣

In the following, we list some fundamental properties of the norms, which can be proved using the properties of inner products.

Proposition 6.2.1

For vectors $\mathbf{v}, \mathbf{w}$ in an inner product space, we have

1. *(Positive-Definiteness)* $\|\mathbf{v}\| \geq 0$ *and* $\|\mathbf{v}\| = 0$ *if and only if* $\mathbf{v} = \mathbf{0}$.
2. *(Homogeneity)* $\|a\mathbf{v}\| = |a| \, \|\mathbf{v}\|$ *where* $a \in \mathbb{F}$.
3. *(Triangle Inequality)* $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$.

♠

Proof Here we only prove the first and the second property, which are immediate results of the properties of inner products. To prove the third property, i.e., the triangle inequality (see Theorem 6.2.3), we need to use the Cauchy-Schwarz Inequality, which will be shown later.

It is clear from the definition of norms that $\|\mathbf{v}\| \geq 0$. Moreover,

$$\|\mathbf{v}\| = 0 \iff \langle \mathbf{v}, \mathbf{v} \rangle = 0 \iff \mathbf{v} = \mathbf{0}$$

We now prove the second property. We have

$$\begin{aligned} \|a\mathbf{v}\|^2 &= \langle a\mathbf{v}, a\mathbf{v} \rangle \\ &= \bar{a}a\langle \mathbf{v}, \mathbf{v} \rangle \\ &= |a|^2 \|\mathbf{v}\|^2 \end{aligned}$$

Hence, $\|a\mathbf{v}\| = |a| \, \|\mathbf{v}\|$. ■

Definition 6.2.2

Two vectors $\mathbf{u}$ *and* $\mathbf{v}$ *in an inner product space are said to be* ***orthogonal*** *to each other if*

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0$$

♣

Theorem 6.2.1 (Pythagoean Theorem)

Let $\mathbf{u}$ *and* $\mathbf{v}$ *be two orthogonal vectors in an inner product space* V*. Then we have the inequality:*

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \tag{6.1}$$

♡

Proof We have

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= \|\mathbf{u}\|^2 + 2\,\mathrm{Re}\,\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2 \end{aligned} \tag{6.2}$$

Because $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, we have $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. Then (6.2) reduces to (6.1). ■

Note *We are very familiar with the equality*

$$(a + b)^2 = a^2 + 2ab + b^2$$

which resembles (6.2) *except that the middle term on the right-hand side of* (6.2) *is taken the real part.*

Since we also consider the complex vector space, we cannot change the order of two vectors in an inner product at will, that is, $\langle \mathbf{u}, \mathbf{v} \rangle \neq \langle \mathbf{v}, \mathbf{u} \rangle$ *in general.*

Next, we shall present the Cauchy-Schwarz Inequality, which is a very important inequality in mathematics. This inequality has many forms across different fields of mathematics, some of which may appear as long scary expressions. In linear algebra, the Cauchy-Schwarz inequality can be simply written as

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|$$

The general idea of its proof is by considering the projection of one vector onto the other one. Suppose $\mathbf{u} \neq \mathbf{0}$, we define a linear map

$$\mathrm{proj}_{\mathbf{u}}\mathbf{v} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|^2}\mathbf{u} \tag{6.3}$$

It is an exercise (Exercise 6.1) to verify that $\mathrm{proj}_{\mathbf{u}}\mathbf{v}$ is indeed a linear map. As the notation suggests, $\mathrm{proj}_{\mathbf{u}}\mathbf{v}$ is the projected vector of $\mathbf{v}$ onto $\mathbf{u}$. See Figure 6.1

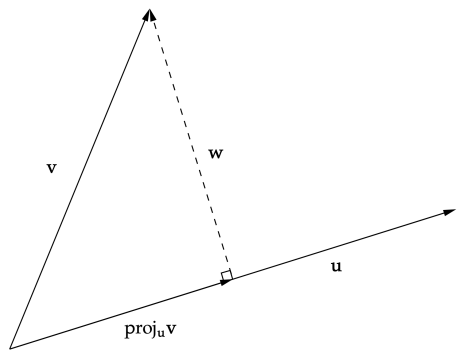

Figure 6.1: Projection of one vector onto another.

Though it is intuitive to see the map $\mathrm{proj}_{\mathbf{u}}$ is a *projection*, we cannot claim so for the time being. Recall we have a formal definition for projections (Definition 2.2.1). That is, a projection P is a linear map satisfying $P^2 = P$. We now show that $\mathrm{proj}_{\mathbf{u}}$ indeed satisfies this requirement. We have

$$\begin{aligned}
\mathrm{proj}_{\mathbf{u}}^2\mathbf{v} &= \mathrm{proj}_{\mathbf{u}}\left(\frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|^2}\mathbf{u}\right) \\
&= \frac{1}{\|\mathbf{u}\|^2}\left\langle \mathbf{u}, \frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|^2}\mathbf{u}\right\rangle \mathbf{u} \\
&= \frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|^4}\langle \mathbf{u}, \mathbf{u}\rangle \mathbf{u} \\
&= \frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|^2}\mathbf{u} \\
&= \mathrm{proj}_{\mathbf{u}}\mathbf{v}
\end{aligned}$$

Therefore, indeed $\mathrm{proj}_{\mathbf{u}}^2 = \mathrm{proj}_{\mathbf{u}}$, i.e., $\mathrm{proj}_{\mathbf{u}}$ is a projection.

Note that $\mathbf{v}$ can be written as

$$\mathbf{v} = \mathrm{proj}_{\mathbf{u}}\mathbf{v} + (\mathbf{v} - \mathrm{proj}_{\mathbf{u}}\mathbf{v})$$

We have

$$\begin{aligned}
\langle \mathrm{proj}_{\mathbf{u}}\mathbf{v}, \mathbf{v} - \mathrm{proj}_{\mathbf{u}}\mathbf{v}\rangle &= \langle \mathrm{proj}_{\mathbf{u}}\mathbf{v}, \mathbf{v}\rangle - \langle \mathrm{proj}_{\mathbf{u}}\mathbf{v}, \mathrm{proj}_{\mathbf{u}}\mathbf{v}\rangle \\
&= \frac{\overline{\langle \mathbf{u}, \mathbf{v}\rangle}}{\|\mathbf{u}\|^2}\langle \mathbf{u}, \mathbf{v}\rangle - \frac{\overline{\langle \mathbf{u}, \mathbf{v}\rangle}}{\|\mathbf{u}\|^2}\frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|^2}\langle \mathbf{u}, \mathbf{u}\rangle \\
&= \frac{|\langle \mathbf{u}, \mathbf{v}\rangle|^2}{\|\mathbf{u}\|^2} - \frac{|\langle \mathbf{u}, \mathbf{v}\rangle|^2}{\|\mathbf{u}\|^2} \\
&= 0
\end{aligned}$$

That is,

$$\mathrm{proj}_{\mathbf{u}}\mathbf{v} \perp (\mathbf{v} - \mathrm{proj}_{\mathbf{u}}\mathbf{v}) \tag{6.4}$$

This implies that $\mathbf{v}$ can be decomposed as a sum of two orthogonal vectors, one of which is its projection onto $\mathbf{u}$. As illustrated in Figure 6.1, $\mathrm{proj}_{\mathbf{u}}\mathbf{v} \perp \mathbf{w}$ where $\mathbf{w} = \mathbf{v} - \mathrm{proj}_{\mathbf{u}}\mathbf{v}$.

From Figure 6.1, we observe that the length of the projected vector is less than or equal to that

of the original one, i.e., $\|\mathrm{proj}_{\mathbf{u}}\mathbf{v}\| \leq \|\mathbf{v}\|$. This is precisely the idea of the proof of Cauchy-Schwarz inequality.

✍ **Exercise 6.1** Show that $\mathrm{proj}_{\mathbf{u}}\mathbf{v}$ defined in (6.3) is a linear map. What if we replace its definition with $(\langle \mathbf{v}, \mathbf{u}\rangle / \|\mathbf{u}\|^2)\mathbf{u}$? Is it still a linear map?

Theorem 6.2.2 (Cauchy-Schwarz Inequality)

Let $\mathbf{u}$ *and* $\mathbf{v}$ *be two vectors in an inner product space. Then*

$$|\langle \mathbf{u}, \mathbf{v}\rangle| \leq \|\mathbf{u}\|\,\|\mathbf{v}\| \tag{6.5}$$

The equality in (6.5) *holds if and only if* $\mathbf{u}$ *and* $\mathbf{v}$ *are linearly dependent.* ♡

Proof If $\mathbf{u} = \mathbf{0}$, then the conclusion is trivial since the equality of (6.5) holds and $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

In the following, we assume $\mathbf{u} \neq \mathbf{0}$. Note that $\mathbf{v}$ can be written as

$$\mathbf{v} = \mathrm{proj}_{\mathbf{u}}\mathbf{v} + (\mathbf{v} - \mathrm{proj}_{\mathbf{u}}\mathbf{v})$$

We have shown in (6.4) that $\mathbf{v} \perp (\mathbf{v} - \mathrm{proj}_{\mathbf{u}}\mathbf{v})$. Then the Pythagorean Theorem (Theorem 6.2.1) implies

$$\|\mathbf{v}\|^2 = \|\mathrm{proj}_{\mathbf{u}}\mathbf{v}\|^2 + \|\mathbf{v} - \mathrm{proj}_{\mathbf{u}}\mathbf{v}\|^2 \geq \|\mathrm{proj}_{\mathbf{u}}\mathbf{v}\|^2 = \frac{|\langle \mathbf{u}, \mathbf{v}\rangle|^2}{\|\mathbf{u}\|^2} \tag{6.6}$$

Rearranging the terms in (6.6) and then taking the square root gives

$$|\langle \mathbf{u}, \mathbf{v}\rangle| \leq \|\mathbf{u}\|\,\|\mathbf{v}\| \tag{6.7}$$

Note that the equality in (6.7) holds if and only if the equality in (6.6) holds. And that happens if and only if

$$\mathbf{v} - \mathrm{proj}_{\mathbf{u}}\mathbf{v} = \mathbf{0}$$

which is equivalent to that $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent. This completes the proof. ■

The famous triangle inequality can be proved using the Cauchy-Schwarz inequality.

Theorem 6.2.3 (Triangle Inequality)

For $\mathbf{u}, \mathbf{v}$ *in an inner product space, we have the following inequality:*

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\| \tag{6.8}$$

The equality in (6.8) *holds if and only if one vector is a* ***nonnegative*** *multiple of the other.* ♡

Proof If $\mathbf{u} = \mathbf{0}$, then (6.8) clearly holds since both sides of are equal to $\|\mathbf{v}\|$. And in this case, we can say $\mathbf{u}$ is a nonnegative multiple of $\mathbf{v}$ by writing $\mathbf{u} = 0\mathbf{v}$.

We now assume $\mathbf{u} \neq 0$ in the rest of the proof. We have

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \|\mathbf{u}\|^2 + 2\,\mathrm{Re}\,\langle \mathbf{u}, \mathbf{v}\rangle + \|\mathbf{v}\|^2 \\ &\leq \|\mathbf{u}\|^2 + 2\,|\langle \mathbf{u}, \mathbf{v}\rangle| + \|\mathbf{v}\|^2 \end{aligned} \tag{6.9}$$

Applying the Cauchy-Schwarz inequality (Theorem 6.2.2) to the right-hand side of (6.9) gives

$$\begin{aligned} \|\mathbf{u}\|^2 + 2\,|\langle \mathbf{u}, \mathbf{v}\rangle| + \|\mathbf{v}\|^2 &\leq \|\mathbf{u}\|^2 + 2\,|\mathbf{u}, \mathbf{v}| + \|\mathbf{v}\|^2 \\ &= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2 \end{aligned} \tag{6.10}$$

Inequality (6.8) then follows by combining (6.9) and (6.10).

Now, we discuss when the equality holds in (6.8). It is clear from the context that the equality in (6.8) holds if and only if the equalities hold simultaneously in (6.9) and (6.10). We know that the equality in (6.9) holds if and only if $\langle \mathbf{u}, \mathbf{v} \rangle \geq 0$ (this implicitly requires that the inner product is a real number). On the other hand, as stated in Theorem 6.2.2, the equality in (6.10) holds if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent. Since $\mathbf{u} \neq \mathbf{0}$, this is equivalent to that $\mathbf{v} = a\mathbf{u}$ for some $a \in \mathbb{F}$. In summary, the equality in (6.8) holds if and only if

$$\begin{cases} \langle \mathbf{u}, \mathbf{v} \rangle \geq 0 \\ \exists a \in \mathbb{F},\ \mathbf{v} = a\mathbf{u} \end{cases}$$

This is equivalent to saying that $\mathbf{v}$ is a nonnegative multiple of $\mathbf{u}$. ■

6.3 Orthonormal Bases

A list of vectors $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ is called **orthonormal** if

$$\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

That is, each pair of vectors in the list is orthogonal and every vector itself is a unit vector.

Exercise 6.2 Suppose $V = \mathbb{R}^3$. What is the norm of the vector $3\mathbf{e}_1 + 4\mathbf{e}_2 + 12\mathbf{e}_3$ where $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ is the standard basis of $\mathbb{R}^3$?

Solution *By definition, squaring the norm in question gives*

$$\begin{aligned} \langle 3\mathbf{e}_1 + 4\mathbf{e}_2 + 12\mathbf{e}_3, 3\mathbf{e}_1 + 4\mathbf{e}_2 + 12\mathbf{e}_3 \rangle &= 3^2 \langle \mathbf{e}_1, \mathbf{e}_1 \rangle + 3 \times 4 \langle \mathbf{e}_1, \mathbf{e}_2 \rangle + 3 \times 12 \langle \mathbf{e}_1, \mathbf{e}_3 \rangle \\ &\quad + 4 \times 3 \langle \mathbf{e}_2, \mathbf{e}_1 \rangle + 4^2 \langle \mathbf{e}_2, \mathbf{e}_2 \rangle + 4 \times 12 \langle \mathbf{e}_2, \mathbf{e}_3 \rangle \\ &\quad + 12 \times 3 \langle \mathbf{e}_3, \mathbf{e}_1 \rangle + 12 \times 4 \langle \mathbf{e}_3, \mathbf{e}_2 \rangle + 12^2 \langle \mathbf{e}_3, \mathbf{e}_3 \rangle \\ &= 3^2 + 4^2 + 12^2 \\ &= 13^2 \end{aligned}$$

Hence, the norm is 13.

Geometrically, one can also argue by the Pythagorean theorem that $3\mathbf{e}_1 + 4\mathbf{e}_2 = (3, 4, 0)^\top$ *is a vector in a* 2*-dimensional plane with norm* 5*. Since* $12\mathbf{e}_3 = (0, 0, 12)^\top$ *is orthogonal to* $(3, 4, 0)^\top$*, we can then apply the Pythagorean theorem again to obtain the final result.*

The second solution to the preceding exercise reveals a deeper meaning of this question and inspires us to generalize the Pythagorean theorem as shown in the next proposition.

Proposition 6.3.1

If $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ *is a list of orthonormal vectors in an inner product space, then*

$$\|a_1\mathbf{e}_1 + \cdots + a_m\mathbf{e}_m\|^2 = |a_1|^2 + \cdots + |a_m|^2$$

where $a_1, \ldots, a_m \in \mathbb{F}$.

Proof We shall prove by induction on m.

(Base Case: $m = 1$) In this case, by the properties of norms we have

$$\|a_1\mathbf{e}_1\| = |a_1|\,\|\mathbf{e}_1\| = |a_1|$$

(Inductive Step) Assume this proposition holds for $m = k$ orthonormal vectors, we need to show it also holds for $m = k + 1$. Note that

$$\langle a_{k+1}\mathbf{e}_{k+1}, a_1\mathbf{e}_1 + \cdots + a_k\mathbf{e}_k\rangle = \sum_{j=1}^{k} \bar{a}_{k+1}a_j\langle \mathbf{e}_{k+1}, \mathbf{e}_j\rangle = 0$$

which means that $a_1\mathbf{e}_1 + \cdots + a_k\mathbf{e}_k$ and $a_{k+1}\mathbf{e}_{k+1}$ are orthogonal to each other. Then by Pythagorean theorem (Theorem 6.2.1), we have

$$\begin{aligned}\|a_1\mathbf{e}_1 + \cdots + a_k\mathbf{e}_k + a_{k+1}\mathbf{e}_{k+1}\|^2 &= \|a_1\mathbf{e}_1 + \cdots + a_k\mathbf{e}_k\|^2 + \|a_{k+1}\mathbf{e}_{k+1}\|^2 \\ &= \|a_1\mathbf{e}_1 + \cdots + a_k\mathbf{e}_k\|^2 + |a_{k+1}|^2\end{aligned}$$

where the last equality follows from the properties of norms. We then apply the induction hypothesis to the vectors $(\mathbf{e}_1, \ldots, \mathbf{e}_k)$. It follows that

$$\begin{aligned}\|a_1\mathbf{e}_1 + \cdots + a_k\mathbf{e}_k + a_{k+1}\mathbf{e}_{k+1}\|^2 &= \|a_1\mathbf{e}_1 + \cdots + a_k\mathbf{e}_k\|^2 + |a_{k+1}|^2 \\ &= |a_1|^2 + \cdots + |a_k|^2 + |a_{k+1}|^2\end{aligned}$$

This completes the proof. ■

The following corollary is a simple application of Proposition 6.3.1, which says a list of nonzero orthogonal vectors must be linearly independent.

Corollary 6.3.1

Every list of orthogonal nonzero vectors $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ in an inner product space is linearly independent. ♡

Proof The equation

$$a_1\mathbf{v}_1 + \cdots a_m\mathbf{v}_m = \mathbf{0}$$

yields

$$\|a_1\mathbf{v}_1 + \cdots a_m\mathbf{v}_m\|^2 = 0$$

We can then compute the square of the norm using Proposition 6.3.1:

$$|a_1|^2\,\|\mathbf{v}_1\|^2 + \cdots + |a_m|^2\,\|\mathbf{v}_m\|^2 = 0$$

This implies

$$a_1 = \cdots = a_m = 0$$

Hence, $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent. ■

In general, it is usually hard to find the components of a vector, $\mathbf{v}$, with respect to some basis. However, the given basis, $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$, is orthonormal, then the component of $\mathbf{v}$ with respect to $\mathbf{e}_j$ is simply the inner product, $\langle \mathbf{e}_j, \mathbf{v}\rangle$.

Theorem 6.3.1

Suppose $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ *is an orthonormal basis of* V*. Then for every* $\mathbf{v} \in V$*, we have*

$$\mathbf{v} = \langle \mathbf{e}_1, \mathbf{v} \rangle \mathbf{e}_1 + \cdots + \langle \mathbf{e}_n, \mathbf{v} \rangle \mathbf{e}_n \tag{6.11}$$

and

$$\|\mathbf{v}\|^2 = |\langle \mathbf{e}_1, \mathbf{v} \rangle|^2 + \cdots + |\langle \mathbf{e}_n, \mathbf{v} \rangle|^2 \tag{6.12}$$

♡

Proof Suppose $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{e}_j$. Taking the inner product with $\mathbf{e}_i$ gives

$$\langle \mathbf{e}_i, \mathbf{v} \rangle = \sum_{j=1}^{n} a_j \langle \mathbf{e}_i, \mathbf{e}_j \rangle = a_i$$

This proves (6.11). And (6.12) follows directly from Proposition 6.3.1 ■

6.4 Gram-Schmidt Process

We appreciate all the benefits and good properties orthonormal bases can bring us. But how can we construct such an orthonormal basis? Suppose we have a list of linearly independent vectors, then we can apply the **Gram-Schmidt process**, which will be introduced in the next theorem, to construct an orthonormal list of vectors out it with the same span.

Theorem 6.4.1 (Gram-Schmidt)

Let V *be a finite-dimensional inner product space over* $\mathbb{F}$*. Suppose* $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ *is a list of linearly independent vectors in* V*, then there exists a list of* ***orthonormal*** *vectors* $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ *in* V *such that*

$$\operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_j) = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_j) \quad \forall 1 \leq j \leq m \tag{6.13}$$

♡

Proof We shall prove by induction on m.

(Base Case) If $m = 1$, then the theorem holds trivially by letting $\mathbf{e}_1 = \frac{1}{\|\mathbf{v}_1\|}\mathbf{v}_1$.

(Inductive Step) Assume that this theorem holds for $m = k$, we shall show that it also holds for $m = k + 1$. Let $(\mathbf{v}_1, \ldots, \mathbf{v}_{k+1})$ be a list of $k + 1$ linearly independent vectors in V. Applying the induction hypothesis to the first k vectors, it follows that there exists as a list of orthonormal vectors $(\mathbf{e}_1, \ldots, \mathbf{e}_k)$ such that

$$\operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_j) = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_j) \quad \forall 1 \leq j \leq k \tag{6.14}$$

We now construct the vector $\mathbf{e}_{k+1}$. Define

$$\mathbf{u} = \mathbf{v}_{k+1} - \sum_{j=1}^{k} \langle \mathbf{e}_j, \mathbf{v}_{k+1} \rangle \mathbf{e}_j \tag{6.15}$$

Note that $\mathbf{u} \neq \mathbf{0}$. Assume $\mathbf{u} = \mathbf{0}$, then (6.15) would imply that

$$\mathbf{v}_{k+1} \in \operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_k) = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$$

which contradicts that $(\mathbf{v}_1, \ldots, \mathbf{v}_{k+1})$ are linearly independent. Now, it is safe to define

$$\mathbf{e}_{k+1} = \frac{1}{\|\mathbf{u}\|}\mathbf{u} \tag{6.16}$$

It is clear $\|\mathbf{e}_{k+1}\| = 1$. We then need to verify that $\mathbf{e}_{k+1}$ is orthogonal to each $\mathbf{e}_j$ $(1 \leq j \leq k)$. It follows from (6.15) and (6.16) that

$$\langle \mathbf{e}_j, \mathbf{e}_{k+1} \rangle = \frac{1}{\|\mathbf{u}\|}\left(\langle \mathbf{e}_j, \mathbf{v}_{k+1} \rangle - \langle \mathbf{e}_j, \mathbf{v}_{k+1} \rangle \cdot 1\right) = 0 \quad \forall 1 \leq j \leq k$$

Therefore, $(\mathbf{e}_1, \ldots, \mathbf{e}_{k+1})$ is indeed a list of orthonormal vectors.

We also need to show (6.13) is satisfied for $m = k+1$. Note that (6.15) and (6.16) imply that $\mathbf{v}_{k+1} \in \operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_{k+1})$. On the other hand, $\mathbf{e}_{k+1}$ can be written as a linear combination of $\mathbf{e}_1, \ldots, \mathbf{e}_k$ and $\mathbf{v}_{k+1}$. Recall (6.14), and we will note that $\mathbf{e}_{k+1}$ can also be written as a linear combination of $(\mathbf{v}_1, \ldots, \mathbf{v}_{k+1})$. Therefore,

$$\mathbf{v}_{k+1} \in \operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_{k+1}) \qquad\qquad \mathbf{e}_{k+1} \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k+1})$$

Combined with (6.14), we obtain that

$$\operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_j) = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_j) \quad \forall 1 \leq j \leq k+1$$

This completes the proof. ■

As proposed in the proof of Theorem 6.4.1, given a list of linearly independent vectors,

$$\mathbf{v}_1, \ldots, \mathbf{v}_m$$

a list of orthonormal vectors can be found as follows.

$$\begin{aligned}
\mathbf{u}_1 &= \mathbf{v}_1 & \mathbf{e}_1 &= \tfrac{1}{\|\mathbf{u}_1\|}\mathbf{u}_1 \\
\mathbf{u}_2 &= \mathbf{v}_2 - \operatorname{proj}_{\mathbf{e}_1}\mathbf{v}_2 = \mathbf{v}_2 - \langle \mathbf{e}_1, \mathbf{v}_2 \rangle \mathbf{e}_1 & \mathbf{e}_2 &= \tfrac{1}{\|\mathbf{u}_2\|}\mathbf{u}_2 \\
&\vdots & &\vdots \\
\mathbf{u}_m &= \mathbf{v}_m - \textstyle\sum_{j=1}^{m-1} \operatorname{proj}_{\mathbf{e}_j}\mathbf{v}_m = \mathbf{v}_m - \sum_{j=1}^{m-1} \langle \mathbf{e}_j, \mathbf{v}_m \rangle \mathbf{e}_j & \mathbf{e}_m &= \tfrac{1}{\|\mathbf{u}_m\|}\mathbf{u}_m
\end{aligned} \tag{6.17}$$

If $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ forms a basis for V, then the list of vectors $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ given by (6.17) is an orthonormal basis for V.

Given any basis of a finite-dimensional inner product space, we can always apply the Gram-Schmidt procedure to convert this basis to an orthonormal one.

Corollary 6.4.1

Any finite-dimensional inner product space has an orthonormal basis. ♡

Proof Let V be an n-dimensional inner product space, and $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ its basis. Theorem 6.4.1 says there exists a list of orthonormal vectors $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ such that

$$\operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_n) = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_n) = V$$

And we know from Corollary 6.6.1 that $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ is linearly independent. (Or we can conclude it is linearly independent just based on the fact that it spans V with precisely n vectors.) Hence, it indeed forms a basis. ■

Recall Theorem 6.2.2. We know that we can extend a list of linearly independent vectors to a

basis. Now, if we have a list of orthonormal vectors at hand, can we extend it to an *orthonormal* basis? Fortunately, we can do so by performing the Gram-Schmidt process.

Corollary 6.4.2

Every orthonormal list of vectors $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ *in a finite-dimensional inner product space can be extended to an orthonormal basis.* ♡

Proof If $\dim V = m$, then we are already done. Suppose the dimension of V is larger than m, say $m + n$. By Theorem 6.2.2, there exists $\mathbf{v}_1, \ldots, \mathbf{v}_n \in V$ such that

$$(\mathbf{e}_1, \ldots, \mathbf{e}_n, \mathbf{v}_1, \ldots, \mathbf{v}_n)$$

forms a basis of V. Then by applying the Gram-Schmidt procedure (Theorem 6.4.1) to this basis, we obtain an orthogonal list of vectors

$$(\mathbf{e}'_1, \ldots, \mathbf{e}'_n, \mathbf{f}_1, \ldots, \mathbf{f}_n)$$

Since this list of vectors is linearly independent by Corollary 6.6.1, and it contains $m + n = \dim V$ vectors, it must be a basis for V.

What is left to show is that the first n vectors $(\mathbf{e}'_1, \ldots, \mathbf{e}'_n)$ remain unchanged. Referring to (6.17), we have

$$\mathbf{e}'_1 = \mathbf{e}_1$$

Having

$$\mathbf{e}'_i = \mathbf{e}_i \quad \forall i = 1, \ldots, j-1$$

where $j \geq 2$, we can compute using (6.17) that

$$\mathbf{e}'_j = \mathbf{e}_j - \sum_{i=1}^{j-1} \langle \mathbf{e}'_i, \mathbf{e}_j \rangle \mathbf{e}'_i = \mathbf{e}_j - \sum_{i=1}^{j-1} \langle \mathbf{e}_i, \mathbf{e}_j \rangle \mathbf{e}_i = \mathbf{e}_j$$

Therefore, we have

$$\mathbf{e}'_j = \mathbf{e}_j \quad \forall j = 1, \ldots, m$$

This completes the proof. ■

6.5 QR Decomposition

We have studied how to conduct the Gram-Schmidt process to construct an orthonormal basis out of any given basis. In terms of matrices, there is a very important matrix decomposition called the **QR decomposition** that corresponds to the Gram-Schmidt process.

Suppose there is a matrix $A \in \mathbb{F}^{m \times n}$ $(m \geq n)$ with full rank, in this case, rank n. Its column vectors are interpreted as linearly independent vectors in $\mathbb{F}^m$. Note that if $m = n$, then the column vectors form a basis. By conducting the Gram-Schmidt process, or in terms of matrices, we have

$$A = QR$$

where Q is an $m \times n$ matrix whose column vectors are orthonormal and R is an upper-triangular matrix with *positive* diagonal entries. By convention, we use letters Q and R to denote these two matrices,

hence the name QR decomposition. We have special names for matrices like Q provided they are square matrices. We call a *real square* matrix an **orthogonal matrix** if it consists of orthonormal columns. The name is changed to **unitary matrix** if the square matrix is *complex*.

It is clear that A consists of the original list of linearly independent vectors, and Q consists of the desired orthonormal vectors. But what is R made of? We shall soon answer this question. Briefly speaking, it consists of the inner products (see (6.17)) we compute during the Gram-Schmidt process.

In fact, we often have two ways to implement the QR decomposition in practice. One is called the **classical Gram-Schmidt Process**, and the other is called the **modified Gram-Schmidt process**.

6.5.1 Classical Gram-Schmidt Process

The classical Gram-Schmidt process is described in Algorithm 2, which is nothing but an algorithmic representation of (6.17).

Algorithm 2: Classical Gram-Schmidt QR Decomposition

Input: $A \in \mathbb{F}^{m\times n}$ $(m \geq n)$ with $\operatorname{rank} A = n$
Output: $Q \in \mathbb{F}^{m\times n}$, $R \in \mathbb{F}^{n\times n}$

1 initialize $Q \in \mathbb{F}^{m\times n}$;
`// initialize` R `as zeros`
2 $R \leftarrow \mathbf{0}_{n\times n}$;
3 **for** $j = 1, \ldots, n$ **do**
`// p maintains the sum of projections`
4 $\mathbf{p} \leftarrow \mathbf{0}_m$;
5 **for** $i = 1, \ldots, j-1$ **do**
6 $R[i,j] \leftarrow \langle Q[:,i], A[:,j]\rangle$;
7 $\mathbf{p} \leftarrow \mathbf{p} + R[i,j]Q[:,i]$;
8 **end**
9 $\mathbf{u} \leftarrow A[:,j] - \mathbf{p}$;
10 $R[j,j] \leftarrow \|\mathbf{u}\|$;
11 $Q[:,j] \leftarrow \frac{1}{R[j,j]}\mathbf{u}$;
12 **end**

To associate with the notations we are familiar with, write

$$A = \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{bmatrix} \quad \text{and} \quad Q = \begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_n \end{bmatrix}$$

By comparing with (6.17), we note that line 2, 6 and 10 in Algorithm 2 define the entries of $R = (r_{ij})$ in the following way:

$$r_{ij} = \begin{cases} \langle \mathbf{e}_i, \mathbf{v}_j \rangle & i < j \\ \|\mathbf{u}_j\| & i = j \\ 0 & i > j \end{cases}$$

This answers our previous question about what R looks like.

You might worry that Algorithm 2 may fail when executing line 11 if $R[j,j] = 0$. But please be rest assured that this cannot happen since we have already shown in the proof of Theorem 6.4.1 that

$\mathbf{u} \neq 0$, i.e., $R[j,j] \neq 0$. So, is our work done here? Is there anything we have yet to prove? Though Algorithm 2 will not generate any errors, it will always produce matrices Q and R, and Q consists of orthonormal columns, we have not however justified that the product of Q and R is A, i.e., $A = QR$. We now prove this in the following.

In each iteration j, line 11 in Algorithm 2 gives

$$\mathbf{u} = R[j,j]Q[:,j] = r_{jj}\mathbf{e}_j \tag{6.18}$$

Replacing $\mathbf{u}$ using line 4 through line 9 yields

$$\mathbf{u} = A[:,j] - \sum_{i=1}^{j-1} R[i,j]Q[:,i] = \mathbf{v}_j - \sum_{i=1}^{j-1} r_{ij}\mathbf{e}_i \tag{6.19}$$

We then equate the right-hand sides of (6.18) and (6.21):

$$\mathbf{v}_j = \sum_{i=1}^{j} r_{ij}\mathbf{e}_i$$

In terms of matrices, this is equivalent to

$$A[:,j] = QR[:,j]$$

since $r_{ij} = 0$ for $i > j$. Because the above equation holds for every j, by stacking $A[:,j]$ together, we finally obtain $A = QR$ as desired.

We summarize the QR decomposition into the following theorem.

Theorem 6.5.1 (QR Decomposition)

*Let $A \in \mathbb{F}^{m\times n}$ $(m \geq n)$ be a matrix with full rank, i.e., rank n. Then A can be decomposed as a product of a matrix Q consisting of orthonormal columns and an upper triangular matrix R with **positive** entries along the diagonal. That is,*

$$A = QR$$

♡

Now, let us work on an example using the QR decomposition.

✍ **Exercise 6.3** Apply the QR decomposition to the matrix

$$A = \begin{bmatrix} 0 & 4 & 5 \\ 1 & 2 & 3 \\ 0 & 0 & 6 \end{bmatrix}$$

Solution *Suppose* $A = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix}$

(Step 1) $\mathbf{u}_1 = \mathbf{v}_1 = (0,1,0)^\top$. $r_{1,1} = \|\mathbf{u}_1\| = 1$. $\mathbf{e}_1 = \mathbf{u}_1/1 = (0,1,0)^\top$.

(Step 2) $r_{1,2}$ *is computed by*

$$r_{1,2} = \langle \mathbf{e}_1, \mathbf{v}_2 \rangle = (0,1,0)(4,2,0)^\top = 2$$

Subtracting the projections, we have

$$\mathbf{u}_2 = \mathbf{v}_2 - r_{1,2}\mathbf{e}_1 = (4,2,0)^\top - 2 \times (0,1,0)^\top = (4,0,0)^\top$$

Then, $r_{2,2} = \|\mathbf{u}_2\| = 4$, *and* $\mathbf{e}_2 = \mathbf{u}_2/4 = (1,0,0)^\top$.

(Step 3) $r_{1,3}$ and $r_{2,3}$ are computed by

$$r_{1,3} = \langle \mathbf{e}_1, \mathbf{v}_3 \rangle = (0,1,0)(5,3,6)^\top = 3$$
$$r_{2,3} = \langle \mathbf{e}_2, \mathbf{v}_3 \rangle = (1,0,0)(5,3,6)^\top = 5$$

Subtracting the projections, we have

$$\mathbf{u}_3 = \mathbf{v}_3 - (r_{1,3}\mathbf{e}_1 + r_{2,3}\mathbf{e}_2) = (5,3,6)^\top - (3 \times (0,1,0)^\top + 5 \times (1,0,0)^\top) = (0,0,6)^\top$$

Then, $r_{3,3} = \|\mathbf{u}_3\| = 6$, and $\mathbf{e}_3 = \mathbf{u}_3/6 = (0,0,1)^\top$.

In summary, matrices Q and R are given by

$$Q = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \textit{and} \quad R = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

6.5.2 Modified Gram-Schmidt Process

6.6 Upper Triangular Matrices With Respect to Orthonormal Bases

Previously, we introduced the upper triangular matrix representation of an operator T in a general vector space. If a vector space is now upgraded to an inner product space by defining an inner product, and we know that $[T]$ is upper triangular with respect to some basis, then we can always find an *orthonormal* basis with respect to which $[T]$ is upper triangular.

Theorem 6.6.1

Suppose V is a finite-dimensional inner product space, and $T \in \mathcal{L}(V)$ is an operator on V. If there exists some basis B for V with respect to which $[T]_B$ is upper triangular, then there must also exist an orthonormal basis B' such that $[T]_{B'}$ is upper triangular. ♡

Proof Write

$$B = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$$

Since $[T]_B$ is upper triangular, by the equivalent condition in Proposition 5.2.1, we have

$$\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$$

is invariant under T for all $k = 1, \ldots, n$. And using the Gram-Schmidt process (Theorem 6.4.1), we can construct an orthonormal basis

$$B' = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$$

with

$$\operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_k) = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k) \quad \forall k = 1, \ldots, n$$

Therefore, $\operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_k)$ is also invariant under T for each k. Hence, $[T]_{B'}$ is upper triangular by Proposition 5.2.1. ■

Recall that we have shown in Theorem 5.2.2 that every operator on a complex vector space can

be represented as an upper triangular matrix with respect to some basis. By applying Theorem 6.6.1, we can specify that basis to be orthonormal.

Corollary 6.6.1

Let V be a complex finite-dimensional inner product space, and $T \in \mathcal{L}(V)$ an operator. Then $[T]$ is upper triangular with respect to some orthonormal basis of V. ♡

Proof It follows directly from Theorem 5.2.2 and Theorem 6.6.1. ■

6.7 Orthogonal Projections

Definition 6.7.1

*Let U be a vector subspace of an inner product space V, the **orthogonal complement** of U, denoted by, $U^\perp$ is defined as the vector subspace of V consists of all vectors in V that are orthogonal to every vector in U. Formally,*

$$U^\perp := \{\mathbf{v} \in V \mid \langle \mathbf{u}, \mathbf{v} \rangle \ \forall \mathbf{u} \in U\}$$

♣

In the definition above, we claim that $U^\perp$ is a vector subspace. Now, we prove this briefly in the following. Suppose $\mathbf{v}_1, \mathbf{v}_2 \in U^\perp$, then for every $\mathbf{u} \in U$, we have $\langle \mathbf{u}, \mathbf{v}_1 \rangle = \langle \mathbf{u}, \mathbf{v}_1 \rangle = 0$. It follows that $\langle \mathbf{u}, \mathbf{v}_1 - \mathbf{v}_2 \rangle = 0$. Hence, $\mathbf{v}_1 = \mathbf{v}_2 \in U^\perp$. Now, suppose $\mathbf{v} \in U^\perp$. We have $\langle \mathbf{u}, \mathbf{v} \rangle = 0 \ \forall \mathbf{u} \in U$. For any scalar $a \in \mathbb{F}$, it follows that $\langle \mathbf{u}, a\mathbf{v} \rangle = a\langle \mathbf{u}, \mathbf{v} \rangle = 0$. And hence, $a\mathbf{v} \in U^\perp$. Therefore, we see that $U^\perp$ is indeed a vector subspace of V.

We also have the following simple observations:

1. $V^\perp = \{\mathbf{0}\}$
2. $\{\mathbf{0}\}^\perp = V$
3. $U_1, U_2 \leq V,\ U_1 \leq U_2 \implies U_1^\perp \geq U_2^\perp$

In what follows, we shall again confine ourselves to finite-dimensional vector spaces to investigate several important results.

Theorem 6.7.1

Let V be a inner product space, and U a finite-dimensional subspace of V. Then

$$V = U \oplus U^\perp$$

♡

Note *Note that V may have infinite dimension. But its subspace U is assumed finite-dimensional.*

Proof Suppose U is not $\{\mathbf{0}\}$ or V. Otherwise the conclusion is trivial. By Corollary 6.4.1, there is an orthonormal basis $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ for U. For any $\mathbf{v} \in V$, we can write

$$\mathbf{v} = \underbrace{\langle \mathbf{e}_1, \mathbf{v} \rangle \mathbf{e}_1 + \langle \mathbf{e}_m, \mathbf{v} \rangle \mathbf{e}_m}_{\mathbf{u}} + \underbrace{\mathbf{v} - \langle \mathbf{e}_1, \mathbf{v} \rangle \mathbf{e}_1 - \cdots - \langle \mathbf{e}_m, \mathbf{v} \rangle \mathbf{e}_m}_{\mathbf{w}} \tag{6.20}$$

Note *The interpretation of* (6.22) *is that we first choose a vector $\mathbf{u} \in U$ by*

$$\mathbf{u} = \langle \mathbf{e}_1, \mathbf{v} \rangle \mathbf{e}_1 + \langle \mathbf{e}_m, \mathbf{v} \rangle \mathbf{e}_m$$

And then let $\mathbf{w} = \mathbf{v} - \mathbf{u}$. *The construction of* $\mathbf{u}$ *is very clever since in this way,* $\mathbf{w}$ *is orthogonal to* $\mathbf{u}$ *as we will see later.*

Note that $\mathbf{w} \in U^\perp$ since

$$\langle \mathbf{e}_j, \mathbf{w} \rangle = \langle \mathbf{e}_j, \mathbf{v} \rangle - \sum_{k=1}^{m} \langle \mathbf{e}_j, \mathbf{v} \rangle \langle \mathbf{e}_j, \mathbf{e}_k \rangle = 0 \quad \forall j = 1, \ldots, m$$

Hence, (6.22) implies that $V = U + U^\perp$.

We also need to show $V = U \oplus U^\perp$, which can be proved by additionally showing that $U \cap U^\perp = \{\mathbf{0}\}$. Indeed, suppose $\mathbf{v} \in U \cap U^\perp$, then

$$\langle \mathbf{v}, \mathbf{v} \rangle = 0$$

where $\mathbf{v}$ in the first slot can be regarded as a vector coming from U, and the second $\mathbf{v}$ is treated as a vector in $U^\perp$. This completes the proof. ■

If V is finite-dimensional, we can then extend the orthogonal basis of U, $\mathcal{B}_1 = (\mathbf{e}_1, \ldots, \mathbf{e}_m)$, to a basis for V,

$$\mathcal{B} = (\mathbf{e}_1, \ldots, \mathbf{e}_m, \mathbf{f}_1, \ldots, \mathbf{f}_n)$$

through the Gram-Schmidt process (Corollary 6.4.2). It is natural to guess that $\mathcal{B}_2 = (\mathbf{f}_1, \ldots, \mathbf{f}_m)$ forms a basis for the orthogonal complement $U^\perp$. The proof is left as an exercise (Exercise 6.4).

✍ **Exercise 6.4** Suppose V is finite-dimensional in the assumption of Theorem 6.7.1. Show that $\mathcal{B}_2$ forms a basis for $U^\perp$.

Solution *First, we show that* $U^\perp$ *is contained in the span of* $\mathcal{B}_2$. *Let* $\mathbf{w} \in U^\perp$. *By definition we have*

$$\langle \mathbf{e}_j, \mathbf{w} \rangle = 0 \quad \forall j = 1, \ldots, m \tag{6.21}$$

But $\mathbf{w}$ *can be written as a linear combination of vectors in* $\mathcal{B}$

$$\mathbf{w} = a_1\mathbf{e}_1 + \cdots + a_m\mathbf{e}_m + b_1\mathbf{f}_1 + \cdots + b_n\mathbf{f}_n$$

Expanding $\mathbf{w}$ *in* (6.21), *we find*

$$0 = \sum_{k=1}^{m} a_k \langle \mathbf{e}_j, a_k\mathbf{e}_k \rangle + \sum_{k=1}^{n} b_k \langle \mathbf{e}_j, \mathbf{f}_k \rangle = a_j \langle \mathbf{e}_j, \mathbf{e}_j \rangle = a_j \quad \forall j = 1, \ldots, m$$

This shows that $\mathbf{w}$ *actually can represented by a linear combination of vectors in* $\mathcal{B}_2$. *Hence,* $U^\perp \subseteq \operatorname{span} \mathcal{B}_2$.

On the other hand, for any vector $\mathbf{w} \in \operatorname{span} \mathcal{B}_2$, *clearly we have* $\langle \mathbf{e}_j, \mathbf{w} \rangle = 0$ *for each* j. *Therefore, the span of* $\mathcal{B}_2$ *is also contained in* $U^\perp$.

In summary, we have shown

$$U^\perp = \operatorname{span} \mathcal{B}_2$$

Note that $\mathcal{B}_2$ *is orthonormal, which implies* $\mathcal{B}_2$ *is linearly independent. Therefore,* $\mathcal{B}_2$ *is indeed a basis for* $U^\perp$.

Corollary 6.7.1

Let V *be an inner product space, and* U *a finite-dimensional vector subspace. Then*

$$U = (U^\perp)^\perp$$

♡

Proof We first show that $U \subseteq (U^\perp)^\perp$. Let $\mathbf{u} \in U$ For any $\mathbf{w} \in U^\perp$, by definition, we have

$$\langle \mathbf{u}, \mathbf{w} \rangle = 0$$

Equivalently,

$$\langle \mathbf{w}, \mathbf{u} \rangle = 0 \quad \forall \mathbf{w} \in U^\perp$$

Therefore, $\mathbf{u} \in (U^\perp)^\perp$ and hence $U \subseteq (U^\perp)^\perp$.

Note *In the proof of $U \subseteq (U^\perp)^\perp$, we did not use the assumption that U is finite-dimensional nor Theorem 6.7.1. But we will need this theorem in proving $U \supseteq (U^\perp)^\perp$. Actually, it suffices to apply the partial result of Theorem 6.7.1 that $V = U + U^\perp$.*

We also need to show the other direction of inclusion. That is, for $\mathbf{v} \in (U^\perp)^\perp$, we need to show $\mathbf{v} \in U$. By Theorem 6.7.1, $\mathbf{v}$ can be written as

$$\mathbf{v} = \mathbf{u} + \mathbf{w}$$

for some $\mathbf{u} \in U$ and $\mathbf{w} \in U^\perp$. Taking the inner product with $\mathbf{w}$ yields

$$0 = \langle \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{w}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle = 0 + \langle \mathbf{w}, \mathbf{w} \rangle$$

It then follows that $\mathbf{w} = \mathbf{0}$ and hence $\mathbf{v} = \mathbf{u} \in U$, as desired. ■

Recall we have already defined what is a projection in Definition 2.2.1. A linear operator P is a projection if

$$P^2 = P$$

This definition is very general but it is also rather abstract. Now, having the concepts of inner product space and orthogonal complement, we can define a projection on an inner-product space with more intuitive physical meanings.

Let V be an inner product space, and U a *finite-decomposition* subspace. By Theorem 6.7.1, we know that every $\mathbf{v} \in V$ can be uniquely written as

$$\mathbf{v} = \mathbf{u} + \mathbf{w}$$

where $\mathbf{u} \in U$ and $\mathbf{w} \in U^\perp$. Then the projection P_U that maps any vector in V to the subspace U can be simply defined by

$$P_U \mathbf{v} = \mathbf{u}$$

Firstly, observe that it is well-defined since the representation $\mathbf{v} = \mathbf{u} + \mathbf{w}$ is unique. And it is indeed a projection since

$$P_U^2 \mathbf{v} = P_U(P_U \mathbf{v}) = P_U \mathbf{u} = \mathbf{u} = P_U \mathbf{v}$$

The following are several more properties, the proof of which is left as an exercise (Exercise 6.5).

1. $\operatorname{im} P_U = U$
2. $\ker P_U = U^\perp$
3. $\mathbf{v} - P_U \mathbf{v} \in U^\perp \ \forall \mathbf{v} \in V$
4. $\|P_U \mathbf{v}\| \leq \|\mathbf{v}\| \ \forall \mathbf{v} \in V$

Exercise 6.5 Prove the above properties.

Solution

Moreover, suppose that $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ is an orthonormal basis for U, then by (6.22), $P_U \mathbf{v}$ can be

written as

$$P_U\mathbf{v} = \langle \mathbf{e}_1, \mathbf{v}\rangle \mathbf{e}_1 + \cdots \langle \mathbf{e}_m, \mathbf{v}\rangle \mathbf{e}_m \tag{6.22}$$

By comparing the above equation (6.22) with the definition of projection on to a single vector, (6.3), we immediately find the relation

$$P_U\mathbf{v} = \mathrm{proj}_{\mathbf{e}_1}\mathbf{v} + \cdots + \mathrm{proj}_{\mathbf{e}_m}\mathbf{v}$$

Let V be an inner product space, and U a finite-dimensional subspace, the problem of minimizing the distance between a given vector $\mathbf{v} \in V$ and each $\mathbf{u} \in U$ occurs frequently in applications. The solution is shown in the next theorem.

Theorem 6.7.2

Let V be an inner product space, and U a finite-dimensional subspace. Then for each $\mathbf{v} \in V$, we have

$$P_U\mathbf{v} = \arg\min_{\mathbf{u}\in U} \|\mathbf{v} - \mathbf{u}\|$$

♡

Proof Let $\mathbf{v}$ be fixed. Because we have already know $P_U\mathbf{v} \in U$, what is left to prove is the following inequality

$$\|\mathbf{v} - P_U\mathbf{v}\| \leq \|\mathbf{v} - \mathbf{u}\| \quad \forall \mathbf{u} \in U \tag{6.23}$$

Note that

$$\mathbf{v} - \mathbf{u} = \underbrace{\mathbf{v} - P_U\mathbf{v}}_{\in U^\perp} + \underbrace{P_U\mathbf{v} - \mathbf{u}}_{\in U}$$

By the Pythagorean Theorem (Theorem 6.2.1), we have

$$\|\mathbf{v} - \mathbf{u}\|^2 = \|\mathbf{v} - P_U\mathbf{v}\|^2 + \|\mathbf{v} - \mathbf{u}\|^2 \geq \|\mathbf{v} - P_U\mathbf{v}\|^2$$

Taking the square root on both sides yields (6.23). ■

The following examples show that how Theorem 6.7.2 can be applied to approximate a continuous function using a polynomial.

Example 6.1 Consider the vector space of all continuous real-valued functions defined on $[-\pi, \pi]$, $C[-\pi, \pi]$. Let the inner product of any pair of functions f and g in $C[-\pi, \pi]$ be defined by

$$\langle f, g\rangle = \int_{-\pi}^{\pi} f(x)g(x)\,\mathrm{d}x$$

Suppose we want to approximate a target function $f(x)$ using a polynomial with degree no larger than 5. In other words, we want to find a polynomial $p(x)$ in the subspace $P_5(\mathbb{R})$ such that the norm

$$\|f(x) - p(x)\|$$

is minimized.

Then by Theorem 6.7.2 and (6.22), the optimal polynomial should be

$$p(x) = \langle p_1, f\rangle p_1(x) + \cdots + \langle p_6, f\rangle p_6(x) \tag{6.24}$$

where $(p_1, \ldots, p_6)$ is an orthonormal basis for $P_5(\mathbb{R})$.

Note *Note that Theorem 6.7.2 is applicable here because the subspace $P_5(\mathbb{R})$ indeed has finite dimension, namely* 6.

To find such an orthonormal basis, we can apply the Gram-Schmidt process to the basis $(1, x, x^2, x^3, x^4, x^5)$.

Suppose the target function we want to approximate is $\sin x$, then the polynomial calculated using (6.24) is

$$p(x) = 0.98786214x - 0.15527141x^3 + 0.00564312x^5$$

If the reader is familiar with calculus of mathematical analysis, then one might suggest approximating $\sin x$ using its Taylor expanding at $x = 0$,

$$\hat{p}(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

Figure 6.2 plots the graphs of the target function $\sin x$ along with the two polynomial approximations, $p(x)$ and $\hat{p}(x)$. As we can see, the orthogonal projection is a better approximation for it nearly overlaps with the target function.

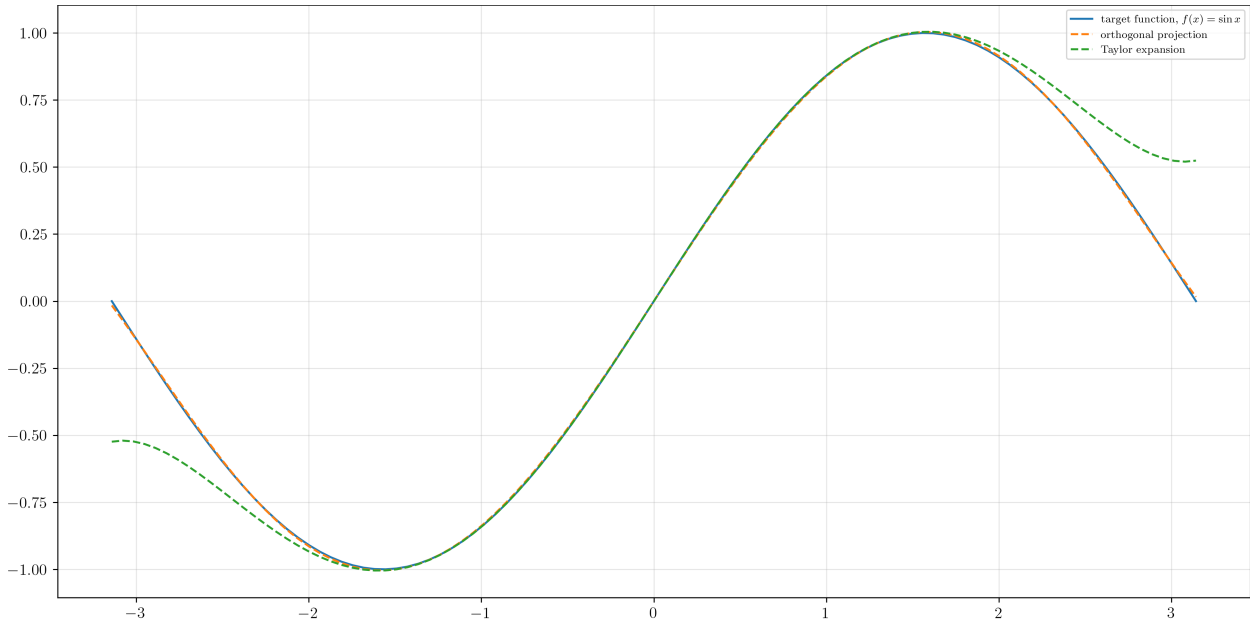

Figure 6.2: Comparison of approximations using orthogonal projection and Taylor expansion.

Example 6.2

6.8 Linear Functionals and Adjoints

Definition 6.8.1

A linear functional, φ, on V is a linear map from V to $\mathbb{F}$,

$$\varphi : V \to \mathbb{F}$$

♣

Recall that the inner product $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ is a bilinear form, and it is linear in the second slot. If we hold the vector $\mathbf{u}$ in the first slot of $\langle \mathbf{u}, \mathbf{v} \rangle$ fixed, then this inner product can be regarded as a linear map from V to $\mathbb{F}$, which is a linear functional. In fact, as we will see in the next theorem, any linear functional on V can be characterized in this form.

Theorem 6.8.1

If φ is a linear functional on a finite-dimensional inner product space V, then there exits a unique vector $\mathbf{u} \in V$ such that

$$\varphi(\mathbf{v}) = \langle \mathbf{u}, \mathbf{v} \rangle \quad \forall \mathbf{v} \in V$$

Note *Note that the inner product space is assumed to have finite dimension.*

Proof We first show the existence of such $\mathbf{u}$. Suppose that $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ is an orthogonal basis of V. Let $\mathbf{v} \in V$ be arbitrary. We can write

$$\mathbf{v} = a_1\mathbf{e}_1 + \cdots a_n\mathbf{e}_n$$

Let $\mathbf{u}$ be given by

$$\mathbf{u} = \overline{\varphi(\mathbf{e}_1)}\mathbf{e}_1 + \cdots + \overline{\varphi(\mathbf{e}_n)}\mathbf{e}_n$$

We have

$$\begin{aligned}
\langle \mathbf{u}, \mathbf{v} \rangle &= \langle \sum_{i=1}^{n} \overline{\varphi(\mathbf{e}_i)}\mathbf{e}_i, \sum_{j=1}^{n} a_j\mathbf{e}_j \rangle \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} \varphi(\mathbf{e}_i)a_j\langle \mathbf{e}_i, \mathbf{e}_j \rangle \\
&= \sum_{i=1}^{n} \varphi(\mathbf{e}_i)a_i \qquad (6.25)
\end{aligned}$$

On the other hand, since φ is linear, it follows that

$$\varphi(\mathbf{v}) = \varphi(\sum_{i=1}^{n} a_i\mathbf{e}_i) = \sum_{i=1}^{n} a_i\varphi(\mathbf{e}_i) \qquad (6.26)$$

Comparing (6.25) and (6.26), we find

$$\langle \mathbf{u}, \mathbf{v} \rangle = \varphi(\mathbf{v})$$

This shows such vector $\mathbf{u}$ is as desired.

Next, we show the uniqueness of $\mathbf{u}$. Suppose

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle \quad \forall \mathbf{v} \in V$$

Then

$$\langle \mathbf{u} - \mathbf{w}, \mathbf{v} \rangle \quad \forall \mathbf{v} \in V$$

In particular, letting $\mathbf{v} = \mathbf{u} - \mathbf{w}$ yields $\mathbf{u} - \mathbf{w} = \mathbf{0}$, which shows that such $\mathbf{u}$ is unique. ■

Definition 6.8.2

Let V and W both be finite-dimensional inner product spaces over field $\mathbb{F}$, and

$$T : V \to W$$

*a linear map. Then the **adjoint** of T, denoted by T^*, is the linear map*

$$T^* : W \to V$$

such that

$$\langle T\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{v}, T^*\mathbf{w}\rangle \quad \forall \mathbf{v} \in V \ \forall \mathbf{w} \in W \tag{6.27}$$

is satisfied. ♣

To show that the adjoint T^* is indeed well-defined, we need to show

1. for each $\mathbf{w} \in W$, there exists a unique vector $T^*\mathbf{w} \in V$ (depending on $\mathbf{w}$) such that (6.27) is satisfied, and
2. T^* is indeed a linear map.

The first statement immediately follows from Theorem 6.8.1. To see this, we first let $\mathbf{w} \in W$ be fixed, and then define a linear functional φ on V by

$$\varphi(\mathbf{v}) = \langle T\mathbf{v}, \mathbf{w}\rangle$$

Note that $\bar{\varphi}(\mathbf{v}) = \overline{\varphi(\mathbf{v})}$ is also a linear function on V. Then by applying Theorem 6.8.1, there exits a unique vector $\mathbf{u} \in V$ such that

$$\bar{\varphi}(\mathbf{v}) = \langle \mathbf{u}, \mathbf{v}\rangle \quad \forall \mathbf{v} \in V$$

Taking the conjugate, equivalently, we have

$$\varphi(\mathbf{v}) = \langle T\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{v}, \mathbf{u}\rangle \quad \forall \mathbf{v} \in V$$

Since $\mathbf{u}$ (uniquely) depends on $\mathbf{w}$, we can write $\mathbf{u}$ as a function of $\mathbf{w}$, $T^*\mathbf{w}$. Hence, (6.27) holds.

We also need to show that T^* is indeed a linear map. For any $\mathbf{w}_1, \mathbf{w}_2 \in W$ and $a, b \in \mathbb{F}$, we have

$$a\langle T\mathbf{v}, \mathbf{w}_1\rangle + b\langle T\mathbf{v}, \mathbf{w}_2\rangle = \langle T\mathbf{v}, a\mathbf{w}_1 + b\mathbf{w}_2\rangle = \langle \mathbf{v}, T^*(a\mathbf{w}_1 + b\mathbf{w}_2)\rangle \quad \forall \mathbf{v} \in V \tag{6.28}$$

Meanwhile, we also have

$$\langle T\mathbf{v}, \mathbf{w}_1\rangle = \langle \mathbf{v}, T^*\mathbf{w}_1\rangle \quad \text{and} \quad \langle T\mathbf{v}, \mathbf{w}_2\rangle = \langle \mathbf{v}, T^*\mathbf{w}_2\rangle \quad \forall \mathbf{v} \in V \tag{6.29}$$

Comparing (6.28) and (6.29), we find that

$$\langle \mathbf{v}, T^*(a\mathbf{w}_1 + b\mathbf{w}_2)\rangle = a\langle \mathbf{v}, T^*\mathbf{w}_1\rangle + b\langle \mathbf{v}, T^*\mathbf{w}_2\rangle = \langle \mathbf{v}, aT^*\mathbf{w}_1 + bT^*\mathbf{w}_2\rangle \quad \forall \mathbf{v} \in V$$

Therefore,

$$T^*(a\mathbf{w}_1 + b\mathbf{w}_2) = aT^*\mathbf{w}_1 + bT^*\mathbf{w}_2$$

which implies T^* is indeed linear.

Example 6.3 The real matrix $A \in \mathbb{R}^{m\times n}$ can be regarded as a linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$. We have

$$\langle A\mathbf{x}, \mathbf{y}\rangle = \mathbf{x}^\top A^\top \mathbf{y} = \langle \mathbf{x}, A^\top \mathbf{y}\rangle$$

Therefore, the adjoint of A is its transpose $A^\top$, i.e., $A^* = A^\top$.

If $A \in \mathbb{C}^{m\times n}$ is a complex matrix, then

$$\langle A\mathbf{x}, \mathbf{y}\rangle = \mathbf{x}^H A^H \mathbf{y} = \langle \mathbf{x}, A^H \mathbf{y}\rangle$$

In this case, the adjoint of A is its Hermitian transpose A^H, or conjugate transpose $\bar{A}^\top$. That is, $A^* = A^H = \bar{A}^\top$.

References

[1] Sheldon Jay Axler. *Linear Algebra Done Right*. 2nd ed. Undergraduate Texts in Mathematics. New York: Springer, 1997. ISBN: 978-0-387-98259-5 978-0-387-98258-8.

Index

A

adjoint of linear map 31

C

Cauchy-Schwarz inequality 17
classical Gram-Schmidt Process 23

G

Gram-Schmidt process 20

I

inner product space 14

M

modified Gram-Schmidt process 23

O

orthogonal 15
orthogonal complement 26
orthogonal matrix 23
orthonormal 18

P

projection 2

Q

QR decomposition 22

U

unitary matrix 23